Question 1 – Real-World Scene Capture to Digital Image Representation

1. Real-World Scene

- A three-dimensional scene contains objects with different shapes, colors, and surface properties.

2. Illumination Source

- A natural or artificial light source illuminates the scene.
- The quality and intensity of illumination affect image appearance.

3. Surface Reflectance

- Object surfaces reflect light based on their material properties.
- Reflection may be diffuse (Lambertian) or specular (mirror-like).

4. Optical Lens System

- The camera lens collects and focuses the reflected light onto the image sensor.
- This forms a continuous image on the imaging plane.

5. Continuous Image Formation

- The scene is represented as a continuous two-dimensional image function f(x,y)f(x, y)f(x,y).
- Image intensity depends on both illumination and surface reflectance.

6. Spatial Sampling

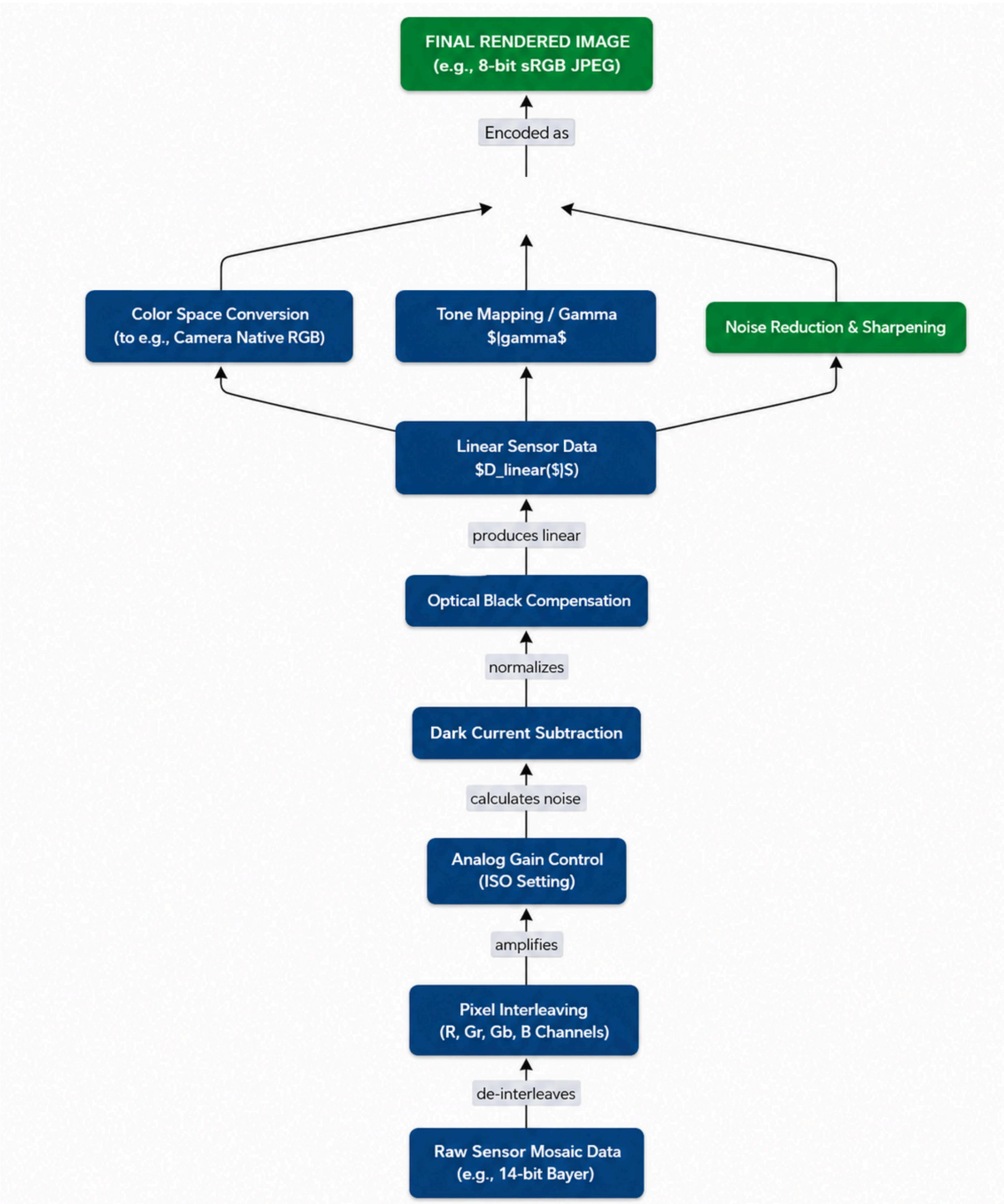
- The CMOS/CCD sensor samples the continuous image into discrete pixels.
- This determines the spatial resolution of the digital image.

7. Amplitude Quantization

- An Analog-to-Digital Converter (ADC) converts the continuous pixel intensity into discrete digital values.

8. Digital Image Representation

- The final output is a digital image matrix I(r,c)I(r, c)I(r,c).
- Each pixel stores a numerical intensity value that can be processed, analyzed, and interpreted by computer vision systems.



Question 2 – Computer Vision Data Handling Pipeline

1. Image Acquisition

- Capture images or videos using a camera, sensor, or imaging device.
- Convert the real-world scene into digital image data.

2. Image Preprocessing

- Improve image quality before analysis.
- Perform operations such as noise removal, contrast enhancement, resizing, normalization, and color correction.

3. Feature Extraction

- Identify important visual characteristics from the image.
- Extract features such as edges, corners, textures, shapes, keypoints, or deep learning features.

4. Feature Analysis

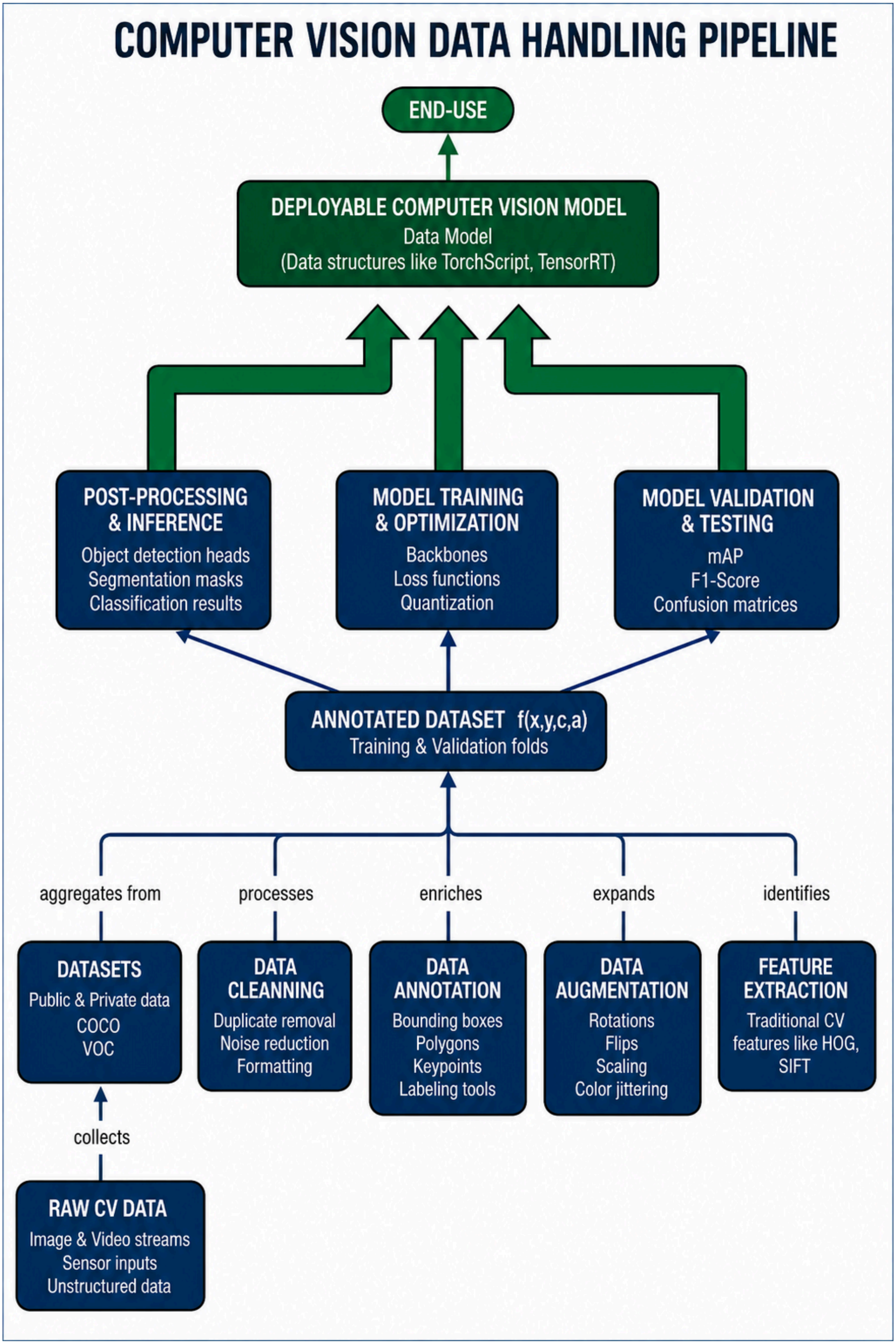
- Analyze the extracted features using Computer Vision algorithms.
- Perform tasks such as image classification, object detection, image segmentation, object tracking, or pattern recognition.

5. Interpretation and Decision Making

- Convert analysis results into meaningful information.
- Recognize objects, understand scenes, make predictions, and support intelligent decision-making for real-world applications.

6. Application Output

- Generate the final output based on the interpreted results.
- Applications include facial recognition, autonomous vehicles, medical image analysis, surveillance, industrial inspection, and robotics.


COMPUTER VISION DATA HANDLING PIPELINE
END-USE
DEPLOYABLE COMPUTER VISION MODEL
Data Model
(Data structures like TorchScript, TensorRT)
POST-PROCESSING & INFERENCE
Object detection heads
Segmentation masks
Classification results
MODEL TRAINING & OPTIMIZATION
Backbones
Loss functions
Quantization
MODEL VALIDATION & TESTING
mAP
F1-Score
Confusion matrices
ANNOTATED DATASET f(x,y,c,a)
Training & Validation folds
aggregates from
processes
enriches
expands
identifies
DATASETS
Public & Private data
COCO
VOC
DATA CLEANNING
Duplicate removal
Noise reduction
Formatting
DATA ANNOTATION
Bounding boxes
Polygons
Keypoints
Labeling tools
DATA AUGMENTATION
Rotations
Flips
Scaling
Color jittering
FEATURE EXTRACTION
Traditional CV features like HOG, SIFT
collects
RAW CV DATA
Image & Video streams
Sensor inputs
Unstructured data

Question 3 – Relationship among Resolution, Spatial Resolution, Intensity Resolution and Image Quality

1. Image Resolution

- Image resolution refers to the total number of pixels in a digital image (width × height).
- Higher image resolution captures more visual detail and improves image clarity.

2. Spatial Resolution

- Spatial resolution represents the size and density of pixels in an image.
- Smaller and more closely packed pixels produce sharper and more detailed images.

3. Intensity Resolution (Bit Depth)

- Intensity resolution indicates the number of brightness or color levels available for each pixel.
- Higher bit depth provides smoother intensity transitions and better color representation.

4. Image Quality

- Image quality depends on the combination of image resolution, spatial resolution, intensity resolution, and noise level.
- Better image quality improves visibility, sharpness, and feature recognition.

5. Impact on Applications

Medical Imaging

- High resolution enables accurate diagnosis.

Satellite Imaging

- Fine spatial resolution improves terrain and object identification.

Face Recognition

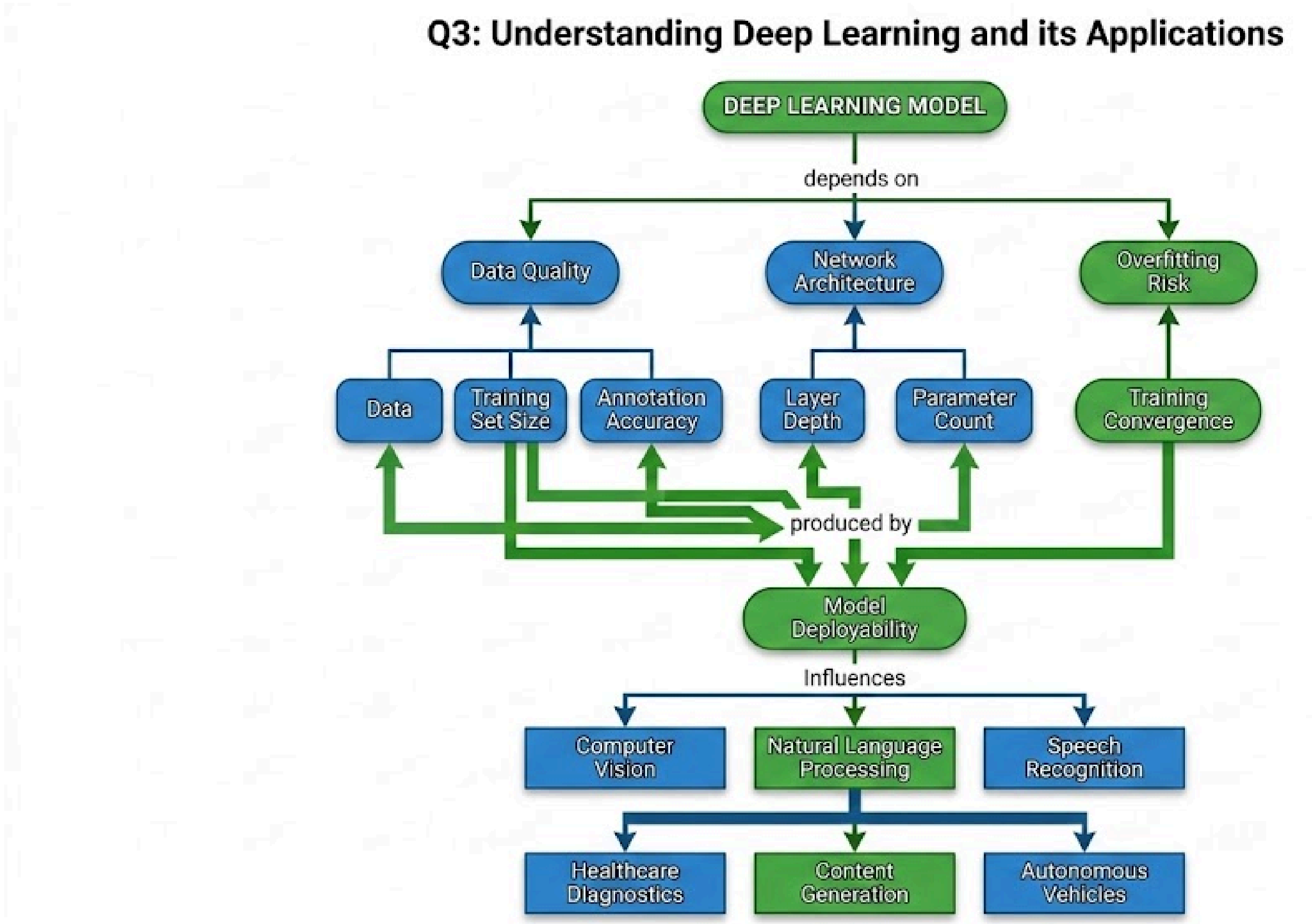
- Better image quality increases recognition accuracy.

Industrial Inspection

- High-quality images help detect defects more precisely.

6. Relationship Among the Factors

- Image resolution, spatial resolution, and intensity resolution are interdependent.
- A balanced combination of these factors produces high-quality images suitable for accurate analysis.



Question 4 – Factors Influencing Image Acquisition Conditions

1. Illumination

- The brightness, direction, and color of the light source affect image appearance.
- Poor lighting can cause shadows, glare, and low contrast, reducing image quality.

2. Sensor Characteristics

- Image sensors (CMOS or CCD) convert light into digital signals.
- Sensor resolution, sensitivity, dynamic range, and noise level determine the quality of the captured image.

3. Motion

- Camera movement or object movement during image capture may produce motion blur.
- Stable cameras and faster shutter speeds help reduce blur and improve image sharpness.

4. Environmental Conditions

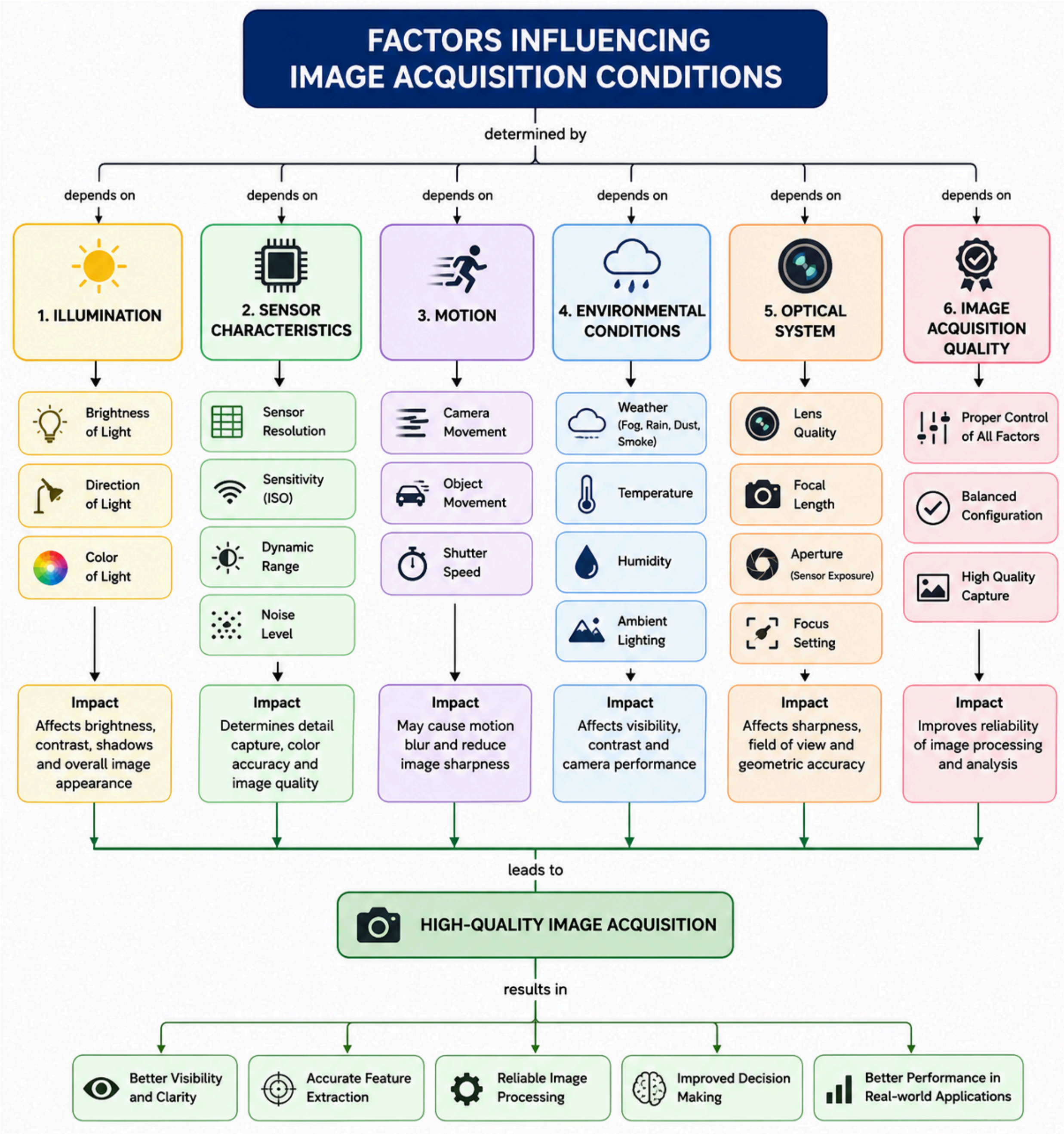
- Weather conditions such as fog, rain, dust, and smoke can reduce image visibility.
- Temperature and humidity may also affect camera performance and image clarity.

5. Optical System

- Lens quality, focal length, aperture, and focus determine image sharpness and field of view.
- Optical distortions and poor focus can reduce image accuracy.

6. Image Acquisition Quality

- Proper control of illumination, sensors, motion, environment, and optics produces high-quality images.
- High-quality images improve the reliability of subsequent image processing tasks.


FACTORS INFLUENCING IMAGE ACQUISITION CONDITIONS
determined by
depends on
1. ILLUMINATION
Brightness of Light
Direction of Light
Color of Light
Impact
Affects brightness, contrast, shadows and overall image appearance
2. SENSOR CHARACTERISTICS
Sensor Resolution
Sensitivity (ISO)
Dynamic Range
Noise Level
Impact
Determines detail capture, color accuracy and image quality
3. MOTION
Camera Movement
Object Movement
Shutter Speed
Impact
May cause motion blur and reduce image sharpness
4. ENVIRONMENTAL CONDITIONS
Weather (Fog, Rain, Dust, Smoke)
Temperature
Humidity
Ambient Lighting
Impact
Affects visibility, contrast and camera performance
5. OPTICAL SYSTEM
Lens Quality
Focal Length
Aperture (Sensor Exposure)
Focus Setting
Impact
Affects sharpness, field of view and geometric accuracy
6. IMAGE ACQUISITION QUALITY
Proper Control of All Factors
Balanced Configuration
High Quality Capture
Impact
Improves reliability of image processing and analysis
leads to
HIGH-QUALITY IMAGE ACQUISITION
results in
Better Visibility and Clarity
Accurate Feature Extraction
Reliable Image Processing
Improved Decision Making
Better Performance in Real-world Applications

Question 5 – Relationship Between Computer Vision System Capabilities and Vision-Based Applications

1. Image Understanding

- The system captures and interprets visual information from images or videos.
- It forms the foundation for all Computer Vision tasks.

2. Object Detection

- Identifies and locates objects within an image.
- Widely used in surveillance, traffic monitoring, and autonomous vehicles.

3. Object Recognition

- Classifies and identifies detected objects based on learned features.
- Used in facial recognition, biometric authentication, and retail systems.

4. Image Segmentation

- Divides an image into meaningful regions or objects.
- Supports medical image analysis, autonomous driving, and satellite image interpretation.

5. Object Tracking

- Continuously follows the movement of objects across video frames.
- Applied in video surveillance, sports analysis, and robotics.

6. Measurement and Inspection

- Measures object size, shape, distance, and detects defects.
- Commonly used in industrial quality control and manufacturing.

7. Decision Making

- Combines outputs from different Computer Vision capabilities to make intelligent decisions.
- Improves automation, accuracy, and efficiency in complex tasks.

8. Vision-Based Applications

- Computer Vision capabilities are applied in healthcare, autonomous vehicles, surveillance, agriculture, robotics, industrial inspection, retail, smart cities, and remote sensing.